

VISUALIZATION: Maximum Heap

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int arr = {14, 33, 52, 23, 12, 65}
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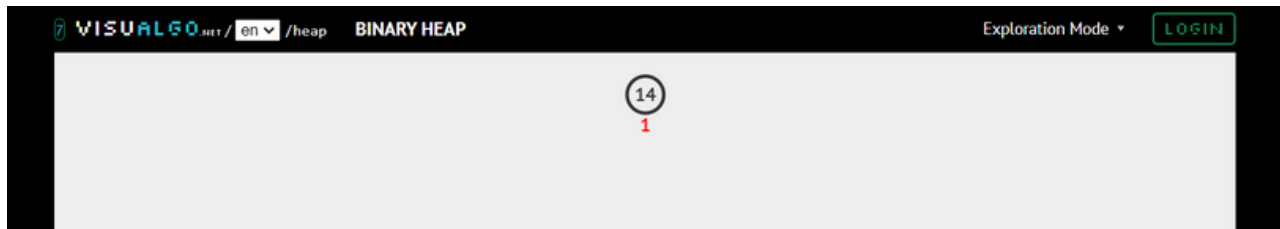
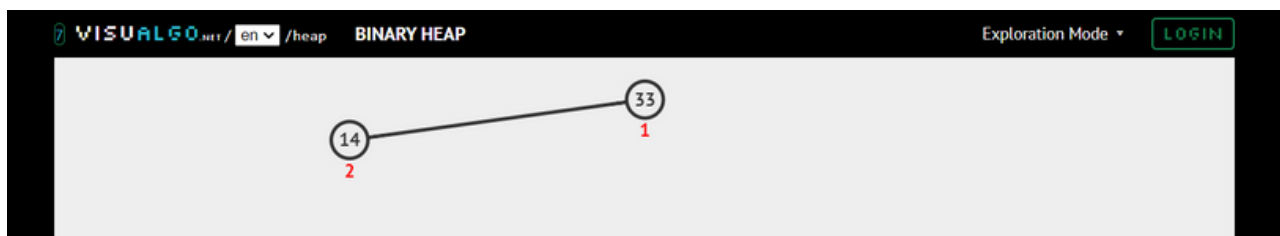
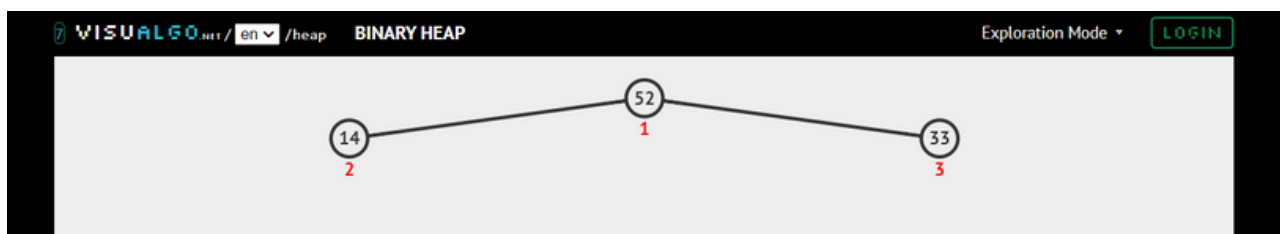
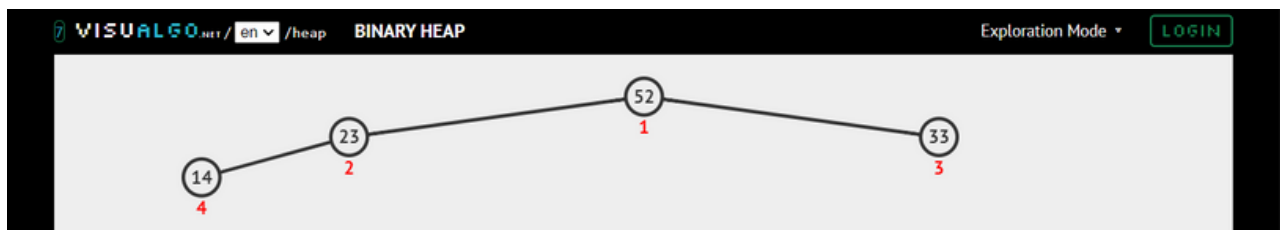


Figure 1. Inserting 14.

Figure 2. When inserting 33, the heap swaps its place from being 14's left child node to its parent node as $33 > 14$.Figure 3. When inserting 52, the heap swaps its place from being 33's right child node to its parent node as $52 > 33$.Figure 4. When inserting 23, the heap swaps its place from being 14's left child node to its parent node as $23 > 14$.

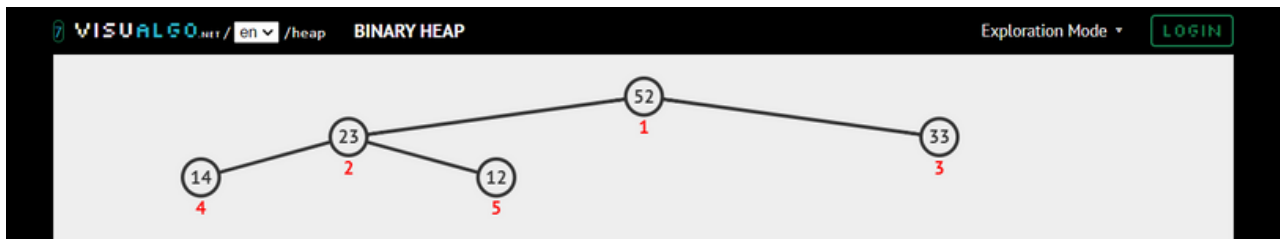


Figure 5. When inserting 12, no swaps occur as $12 < 23$, leaving it as 23's right child node.

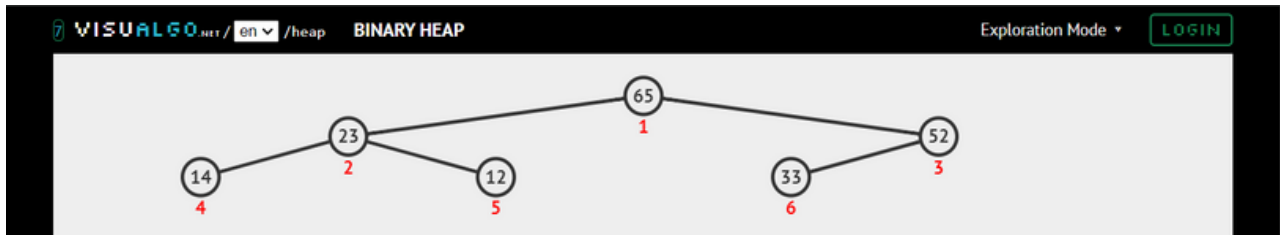


Figure 6. When inserting 65, the heap compares it to 33 and 52 and since $65 > 33$ & 52, it becomes the root node with 52 and 33 having placed in their correct position through the sifting down process.

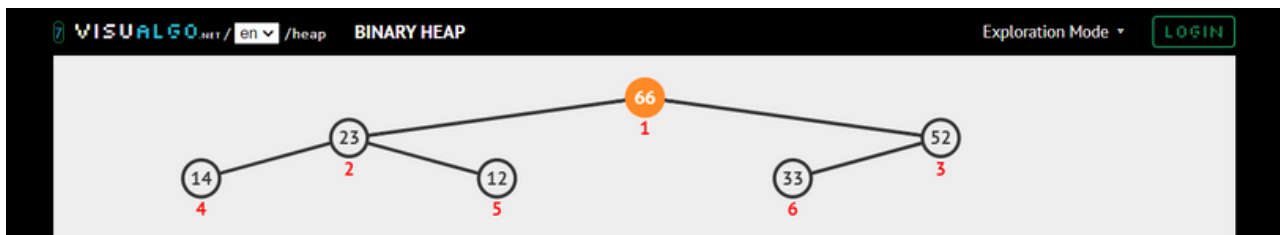


Figure 7. When deleting 65, the heap updates node to be deleted by adding 1 to its previous value.



Figure 8. Then, it is taken out of the heap.

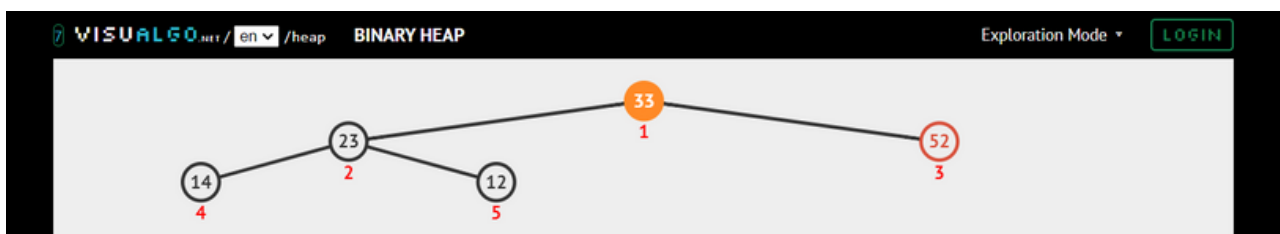


Figure 9. With 33 being the last leaf of the tree, it replaces the root.

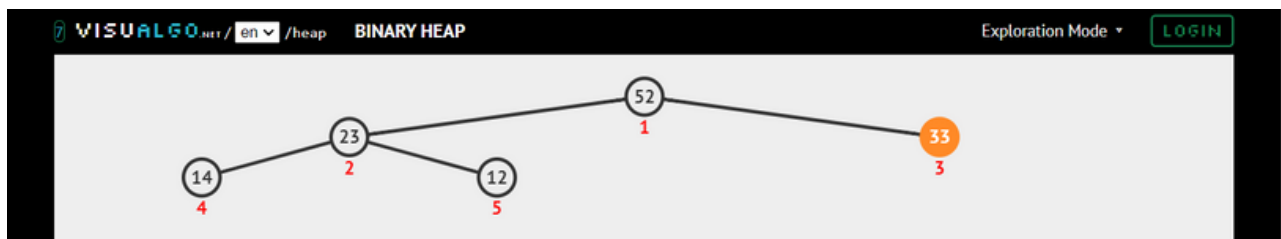


Figure 10. Since $33 < 52$, they swap.

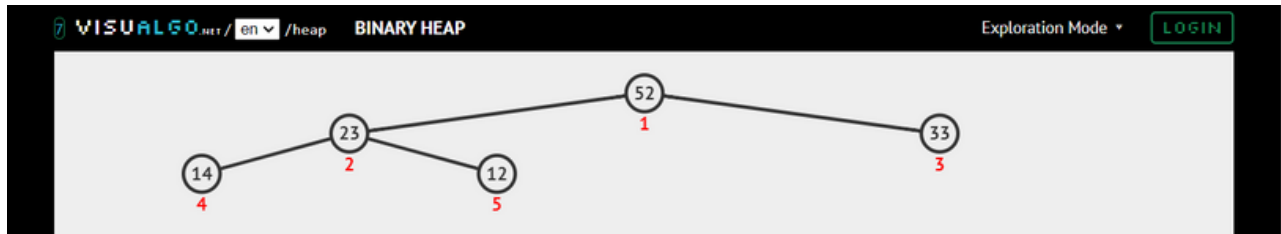


Figure 11. Final maximum heap is 52, 23, 35, 14, and 12.